



SYSTEM VULNERABILITY MANAGEMENT (CYS330)

Understanding logic Gates/Circuits

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LOGIC GATES / CIRCUITS

- These are fundamental building blocks of digital circuits, performing basic logical operations on binary inputs (0s or 1s) to produce a binary output.
- Forming the basics for complex computer systems and electronic devices
- Types
- **AND GATE**
- Performs logical multiplication of binary input.
 - The output state of the AND GATE will be high (1)
 - If both output are high(1) “1 x 1” else the output will be low (0)
 - If any of the input is low(0)

The value of X will be true when both the input will be True

AND GATE Cont'd

- Logical Notations

- $Z = A \text{ AND } B$

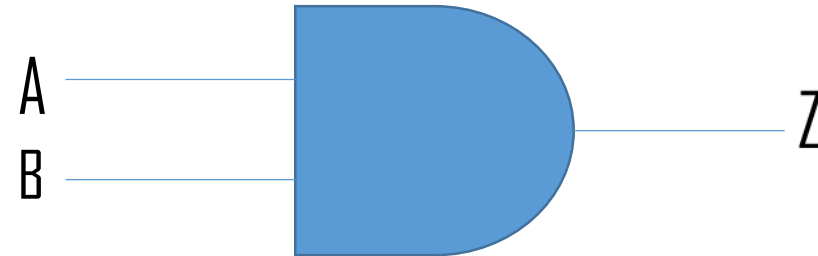
- Boolean algebra

- $Z = A.B$

- A (input 1) B(input 2) $Z=(A.B)$

0	1	0
1	0	0
1	1	1
0	0	0

symbol (x)



OR GATE

- Is the most widely used digital logic circuit
- The output state of OR GATE will be high i.e. (1) if any of output state is high i.e. 1 else output state will be low i.e. 0 X
- Boolean expression
- $X = A + B$
- The value X will be high (true) when one of the inputs is set to high (true)



-
- | Input (A) | input (B) | output (X) |
|-----------|-----------|------------|
| 1 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 1 | 1 |
| 0 | 0 | 0 |

Input (A)	input (B)	output (X)
1	0	1
0	1	1
1	1	1
0	0	0

Logical Notations

$$X = A \text{ OR } B$$

Boolean

$$X = A + B$$

What created the need for NGFWs?

■ NOT GATE

- In digital electronics the NOT GATE is one of the basic logic GATE having only a single input and a single output, Its also known as an inverter or inverting buffer.

- When the input single is **low** the output single is **high** and vice-versa

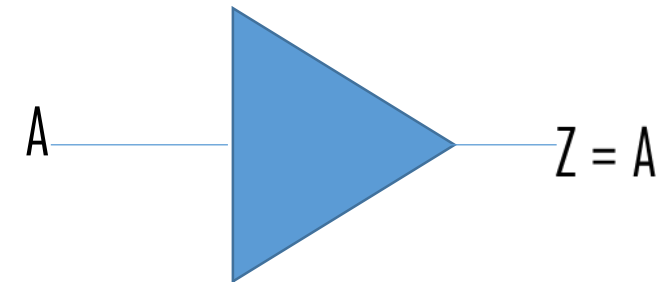
- The Boolean expression of NOT GATE is as follows

- $Y = A'$ or
- $Y = \bar{A}$

- The value of Y will be high when A will be low

- A (input) $Z = A'$ (output)

0	1
1	0



Not Gate takes only one output

the output of a not gate is a complement or inverse of the input applied to it.

HOW TO CONVERT A DECIMAL NUMBER INTO BINARY (POWERS OF 2)

- To achieve this will utilize the powers of 2.

- $2^0 = 1$

- $2^1 = 2$

- $2^2 = 4$

- $2^3 = 8$

- $2^4 = 16$

- $2^5 = 32$

- $2^6 = 64$

- $2^7 = 128$

```
>>> bin(75)
'0b1001011'
>>> _
```

- For example, let's convert 75 into binary

- 128 64 32 16 8 4 2 1

0 1001011 = 75
Sign (+) Magnitude

0 1001011 = 75
Sign (+) Magnitude

HOW TO CONVERT A DECIMAL NUMBER INTO BINARY (POWERS OF 2) (CONTINUES...)

- Let's take another example of 339



- 1 = True
- 0 = False

```
>>> int (True)
1
>>> int (False)
0
>>>
```

```
>>> bin(339)
'0b101010011'
>>> _
```

HOW TO CONVERT A DECIMAL NUMBER INTO BINARY SUCCESSIVE DIVISION

Now, let's use successive division to find the binary of 75.

$$75 / 2 = 37 \text{ R } 1$$

$$37 / 2 = 18 \text{ R } 1$$

$$18 / 2 = 9 \text{ R } 0$$

$$9 / 2 = 4 \text{ R } 1$$

$$4 / 2 = 2 \text{ R } 0$$

$$2 / 2 = 1 \text{ R } 0$$

$$1 / 2 = 0 \text{ R } 1$$

- We now get the binary from the bottom to the top, making it:

- 1001011

```
>>> bin(75)
'0b1001011'
>>> 0b1001011
75
>>>
```


HOW TO ADD BINARY NUMBERS

- Here are some basics of addition:

- $0 + 0 = 0$
- $0 + 1 = 1$
- $1 + 0 = 1$
- $1 + 1 = 10$

$$\begin{array}{r} 1 \\ + 1 \\ \hline \end{array}$$

```
>>> bin(1+1)
'0b10'
>>>
```

1 0 , this is the same as $1 + 1 = 2$

- $2 / 2 = 1 \text{ R } 0$

HOW TO ADD BINARY NUMBERS (CONTINUES...)

1
1
+ 1

```
>>> bin(1+1+1)
'0b11'
>>>
```

- NOTE: On the bottom we are putting the reminder (R) and carrying the number of times 2 enters into the target number

1 1 This is the same as $1+1+1 = 3$

■ $3 / 2 = 1 \text{ R } 1$

HOW TO ADD BINARY NUMBERS (CONTINUES...)

- ■ Let's take an example below:

1010 = 10

+ 1001 = 9

10011 = 19

16	8	4	2	1	Decimal
	1	0	1	0	10
	1	0	0	1	9
1	0	0	1	1	19

```
>>> bin(10)
'0b1010'
>>> bin(9)
'0b1001'
>>> bin(19)
'0b10011'
>>>
```

HOW TO SUBTRACT BINARY NUMBERS

Here are some basics of subtraction:

$$1 - 0 = 1$$

$$1 - 1 = 0$$

```
>>> bin(8-1)
'0b111'
>>> |
```

$$10 \rightarrow 02$$

$$- 1 \rightarrow 1$$

—

	1	1	
0	2	2	2
1	0	0	0
-			1
0	1	1	1

→

8	4	2	1
8			
-			1
			7

1 In base 2, will get a 2 from the nearest value

Binary Number system

understanding overflow with calculations

- A computer uses digits but there is time that a computer switches to octal, decimal and hexadecimal number system.
- Octal and hexadecimal will be skipped you can learn on your free time.
- Will look at Binary Number system
- Addition of Binary Numbers – we have two elements 0s and 1s
- Example 1

$$\begin{array}{r} 1101 \\ + 111 \\ \hline \end{array}$$

1'S & 2'S COMPLEMENT

1'S COMPLEMENT

- Obtain 1's complement of 1010.
- NOTE: In ones complement we flip the numbers, 1 becomes 0 and 0 becomes 1.
- Therefore 1's complement of 1010 is 0101

2'S COMPLEMENT

- Formula:

2's complement = 1's complement + 1

QUESTION: Obtain 1's complement of 10111010.

1s complement = 01000101

2s complement = 01000110

SIGNED MAGNITUDE

CONVERTING A BINARY

- Let's take an example, let's express (-6)

using the signed magnitude.

1. What is (+6) in binary?

8 4 2 1
0 1 1 0 = +6
↓
Sign (+)
→ Magnitude

NEGATIVE BINARY REPRESENTATION

2. The Sign (+) will change from 0 to 1
Sign (-).

8 4 2 1
Sign (-) ← 1 1 1 0

NOTE: 1 represent negative and 0 positive

2'S COMPLEMENT

NEGATIVE NUMBERS

- Represent -12 in binary using the 2's complement.

16 8 4 2 1

0 1 1 0 0 = 12

1 0 0 1 1 = 1's complement

2'S COMPLEMENT (-12)

1 0 0 1 1

+ 1

1 0 1 0 0 = -16 + 4 = -12

Sign (-) Magnitude

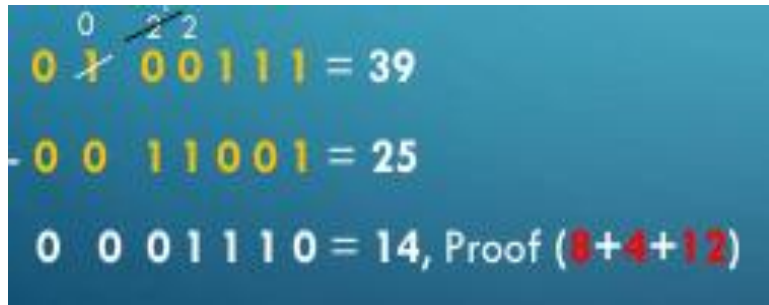
2'S COMPLEMENT SUBTRACTION

BINARY CONVERSION

- Let's subtract $39 - (25)$.
- Using normal math operation $39 - 25 = 14$.

1. Convert 39 and 25 into binary.

64 32 16 8 4 2 1



A binary conversion diagram showing the conversion of decimal numbers 39, 25, and 14 to binary using a place value chart. The chart has columns for powers of 2: 0, 2, 4, 8, 16, 32, 64. The numbers are written in yellow and red. The result 14 is shown as 0001110, with a proof of 8+4+2.

$$\begin{array}{r} 0 \quad 2 \quad 4 \quad 8 \quad 16 \quad 32 \quad 64 \\ 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 = 39 \\ - 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 = 25 \\ \hline 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 = 14, \text{ Proof } (8+4+2) \end{array}$$

1'S & 2'S COMPLEMENT

2. Find the 1's complement 25.

1 1 0 0 1 1 0 = 25 1's complement

3. Find the 2's complement of 25.

1 1 0 0 1 1 1 = 25 2's complement

NOTE: We are targeting **25**, because in 2's complement subtraction we **change the sign** of the number after the minus (-) sign, so **25** becomes -**25**.

2'S COMPLEMENT SUBTRACTION (CONTINUES...)

2'S COMPLEMENT SUBTRACTION

4. Finally, do the 2's complement subtraction of 39 and 25.

$$\begin{array}{r} 0100111 \\ + 1100111 \\ \hline (1)0001110 \end{array}$$

NOTE: We have a carry 1 at the end, but we are just ignoring it, meaning an Overflow as occurred.

VERIFICATION

- Is 0001110 14?

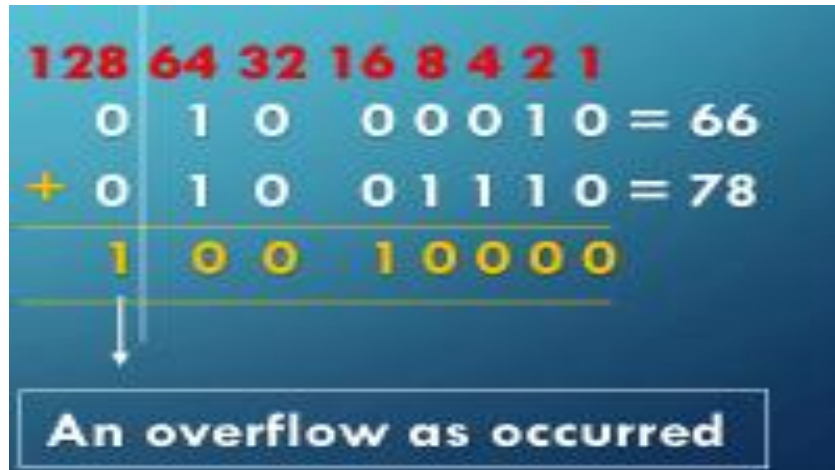
128 64 32 16 8 4 2 1

0 0 0 0 1 1 1 0

- $8 + 4 + 2 = 14$, NOTE: In 2's complement subtraction we add two binaries, think of it as $39 - 25$ is the same as $39 + (-25)$

2'S COMPLEMENT SUBTRACTION (Example 2)

1. $66 - (-78) = 144$
2. This is the same as $66 + (78)$



Intentionally left blank

2'S COMPLEMENT Addition (example 1)

1. -47 and 19

- 47 + 19 or 19 - 47 = -28

19 + (-47)

0	0	0	1	0	0	1	1	= 19
+	1	1	0	1	0	0	0	= 47's 2's comp.
1	1	1	0	0	1	0	0	

\$

128	64	32	16	8	4	2	1	
0	0	0	1	0	0	1	1	= 19
0	0	1	0	1	1	1	1	= 47
1	1	0	1	0	0	0	0	= 47's 1's compliment
+						1		
1	1	0	1	0	0	0	1	= 47's 2's compliment

2'S COMPLEMENT Addition (example 2)

1. 25 and -56

\$

$$25 + (-56) = -31$$

0	0	0	1	1	0	0	1	= 25
+	1	1	0	0	0	1	1	= 56's 1's comp.
1	1	1	0	0	0	0	0	= -31

128	64	32	16	8	4	2	1	
0	0	0	1	1	0	0	1	= 25
0	0	1	1	1	0	0	0	= 56
1	1	0	0	0	1	1	1	= 56's 1's comp.

QUESTIONS ?

